COMSATS University Islamabad,
Lahore Campus

MidTerm Exam □ Terminal Examination – SP20

Course Title:	Game Development		Course Code:	CSC495	Credit: 3
Course Instructor/s:	Saira Aslam, Mohsin Mehdi		Program Name:	BCS/BSE	
Semester:	6th/7th	Batch: FA18-BCS, SP19-BCS, SP19-BSE	Section:	Elective Elective A/B/C	Date: 17th November 2021
Time	I hour 30 minutes		Maximum Marks:		50
Student's	Noor Tariq Rana		Reg. No.	SP19-BSE-005	

Important Instructions / Guidelines:

- Total number of pages are 6
- Make sure to write your Name and Registration number at the top of paper
- Perform your Exam in the space provided with every question
- Do not exceed the space limit

	SECTION 1		SECTION 2		
Question #	1	2	3	4	Total
Obtained Marks	6	4	8.5	10	28.5 46
Total Marks	10	10	20	10	50

SECTION 1 [20 MARKS]

Q1 (a) Elaborate how any one particular game can be different for a player, on a mobile and PC? [2.5+2.5=5]

One particular game can be different for a player on mobile and PC. Firstly there will be a big interface change. Secondly the screen width of mobile and PC vary from each other. In PC we have keyboard to control all our motions but in mobile we have on screen keyboard. Or we have given a control button box on the side of a game interface in mobile to play game but in PC we have keyboard separately.

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(3) It can be detected by onTriggerEnter function or by onCollisionEnter. In onTriggerEnter we can specify if the bullet trigger and hit the enemy it will die, In onCollision we can make the equation (condition) that if the bullet hits the enemy 3 times destroy it.

b) If a game object has a NevMesh Agent as component? How can you make it Navigate on a surface? Write steps. [4]

NavMesh is an component of AI.
If we have NavMesh Agent as a component we can make it navigate on the surface, first we make NavMesh Agent object. NavMesh allow the particular character you want navigate around the ground. It follows the destination which you to follow up.

```
using. UnityEngine.AI;
public navMeshAgent enemy; // we want it to follow the posit
{                          or more it in th
                           position of our play
enemy. destination = player;

}
```

This will make the enemy object to follow the position of what we have given to destination.

Q1 (b) Suppose you want to instantiate 10 cars in a line in Unity Game scene on road. What can be the possible value for coordinates you give for Vector3 pos = new Vector3(_____)?

Also mention how position of each coordinate can be accessed using pos? [2+3=5]

We have to instantiate 10 cars in line in unity Game on road. The possible for the coordinates are,

get.axes("Vertical");
get.axes("Horizontal");
Vector3 pos = Transform.position(x, y, 0)

Vector3 pos = newVector3 (Transform.position (x, y, 0))

1

Q2: Explain the following in Unity:

a) If you have animation clip "EnemyDie",
   1) How can you make it part of an existing Animation Controller?
   2) What Parameter will you set in this animation controller to make the Enemy die?
   3) What possible method in CS script can detect if a bullet hits the enemy and "EnemyDie" can be triggerd? [2+2+2=6]

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(1) We already have an animation clip, we can make it part of an existing animation controller by draging that animation clip into our game & applying it to EnemyDie.

(2) The parameter we will set in this animation to make the enemy die are when the 1st bullet hit the enemy, his remaining energy should be 75%, when second bullet hit the enemy it energy should be 45%, when third bullet hit the enemy it energy should be 0% and that enemy (Game Object) will be destroy. hence the enemydie.

(3) In (OnTriggerEnter) we can specify if the bullets hits the enemy trigger the Enemydie